

Lactose Intolerance Report

LCT

Lactase enzyme

Background science:

Lactose intolerance occurs when the body reduces or stops the production of lactase, an enzyme needed to digest lactose, which is the sugar found in dairy products. Whether an individual produces lactase into adulthood is primarily determined by genetic variants near the LCT gene. People of European ancestry often carry these variants that allow continued lactase production, while the vast majority of those with Asian ancestry do not, making them more likely to experience lactose intolerance as adults. Additionally, certain gut microbes, like *Lactobacillus* species, can help break down small amounts of lactose, meaning some who are genetically lactose intolerant may still tolerate small quantities of dairy without significant discomfort.

The prevalence of lactase persistence or intolerance varies by ancestry due to historical dietary habits. In Europe, dairy farming created a survival advantage for individuals who could digest milk, leading to high rates of lactase persistence. The primary genetic variant associated with this is rs4988235 and is mostly found in people of European descent. Other populations, including those of African and Middle Eastern ancestry, may have different genetic mutations supporting lactase persistence.

Lactose intolerance is not the same as a milk allergy. Lactose intolerance results from enzyme deficiency, while allergies are an immune response to milk proteins.

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
LCT	rs4988235	GG	G	0.49	For GG only: no lactase produced as an adult (lactose intolerant)
MCM6	rs145946881	--	G	0	For GG only: no lactase produced as an adult (African ancestry)
MCM6	rs41380347	--	A	0.99	For AA only: no lactase produced as an adult (Middle Eastern Ancestry)

Understanding your data:

The lactase persistence genetic variants are different, depending on your ancestry.

→ If you have mainly **Caucasian European or Asian ancestry**, read this:

Likely Lactose Intolerant: Your genotype for rs4988235 is **GG**, which means you likely do not produce lactase as an adult. Note that you may still be able to tolerate some dairy products because your gut bacteria can also break down lactose.[\[ref\]](#)

→ If you have mainly **sub-Saharan African ancestry**, read this:

→ If you have mainly **Middle Eastern ancestry**, read this:

References: Please see the full lactose intolerance article here for a full list of references and a more detailed explanation: <https://www.geneticlifehacks.com/lactose-intolerance/>

Disclaimer: This report is for informational and educational purposes only. It is not intended to treat, diagnose, or cure any condition.